Okoro Chukwueloka Nnaemeka

Electronic Engineering · Embedded Systems · Cyber-Physical Power Systems · Edge ML
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RESEARCH PROFILE | MORE ↗

Final-year B.Eng. Electronic Engineering student at the Federal University of Technology Owerri (CGPA 3.0/4.0; expected graduation 5 Dec. 2026) with research interests in **cyber-physical protection and restoration for power distribution**, **embedded firmware on ARM Cortex-M**, and **edge machine learning for fault diagnosis**. First author of a preprint on human-on-the-loop feeder restoration, co-author of a preprint on a portable oscilloscope and programmable power supply, and undergraduate research assistant on tamper-proof IoT telemetry. Applying for the M.Sc. in Electrical and Electronic Engineering under the Chinese Government Scholarship (CSC). Detailed project write-ups: portfolio link above.

EDUCATION | MORE ↗

B.Eng. Electronic Engineering — Expected 5 Dec. 2026
Federal University of Technology Owerri (FUTO), Imo State, Nigeria — *CGPA 3.0/4.0*

- Coursework: Power Electronics, Control Systems, Digital Signal Processing, Microprocessor Systems, Semiconductor Physics, Circuit Analysis, Electromagnetic Fields, Reliability and Quality Assurance.
- Language of instruction: English (100% medium of instruction).

PUBLICATIONS AND PREPRINTS | MORE ↗

- **C. N. Okoro**, "A Human-on-the-Loop Cyber-Physical Framework for Fault-Aware Service Restoration in Nigerian Distribution Feeders: Requirements Derived from EEDC Industrial Training," *engRxiv* preprint, Oct. 2026. DOI: 10.31224/8498. *Research framework and validation protocol; not yet validated on a utility feeder.*
- E. P. Chinonyerem, O. D. Chinazaekpere, C. S. Chinyere, and **C. N. Okoro**, "PowerScope: Preliminary Design of a Portable Dual-Channel Digital Oscilloscope and Programmable DC Power Supply," *engRxiv* preprint, Oct. 2026. DOI: 10.31224/8499.

RESEARCH EXPERIENCE | MORE ↗

Undergraduate Research Assistant — Oct. 2025 – present
Department of Electronic Engineering, FUTO — *Working paper in preparation*
Tamper-proof cyber-physical IoT telemetry for regulatory drinking-water quality compliance (NAFDAC). With Prof. G. Chukwudebe, Prof. U. Ezeh and Dr. C. Anyalewechi.

- Building edge sensor acquisition and hardware-level anti-tamper detection routines for an end-to-end telemetry network with a live supervisory dashboard.
- Co-drafting the System Architecture, Hardware Security and Experimental Validation sections of the manuscript.

Undergraduate Thesis: PowerScope — 2025 – 2026
Lead hardware and firmware designer, FUTO — *Portable dual-channel oscilloscope and programmable supply*

- Designed a dual-channel 10 MS/s oscilloscope front end (14-bit AD9251 ADC) and a 1–15 V, 2 A programmable buck-boost supply (LM5176 controller, LTC4015 charger interface).
- Specified an ESP32-S3 supervisory controller for display, parameter control and wireless streaming, and an Edge Impulse TinyML pipeline for circuit fault classification using DWT and FFT features.
- Modelled the analogue front-end scaling, frequency response and transient behaviour in Proteus VSM.

SELECTED TECHNICAL PROJECTS | MORE ↗

GridPulse-RESTORE: Protection and Human-Approved Restoration Testbed — 2026, in progress
STM32F103 and ESP32, Proteus 8 — *Scaled 24 V / 50 Hz network*

- Designed a three-feeder scaled network with an STM32 protection controller, hardware overcurrent trip latches, break-before-make dead transfer, a capacity check and an 11-state restoration machine. The ESP32 only forwards operator approvals, which the STM32 re-verifies.
- Built the source, earthing, feeder and part of the measurement blocks in Proteus. Calculated fault currents and graded trip times; closed-loop simulation is pending.

Real-Time UAV Motor Monitoring System — June 2025
Group leader, team of 9 — *ATmega, Bluetooth and LoRa*

- Integrated temperature, RPM, fuel-flow and piezoelectric vibration sensors with moving-average and Kalman filtering at under 1 ms latency.
- Deployed an SVM anomaly classifier (F1 = 0.95) and clock-gating power optimisation, keeping consumption below 500 mW at 100 Hz sampling.

VibeGuard: Vibration Monitoring and Live Dashboard 2026
STM32F103, ADXL345, Proteus *UART link to Python dashboard*

- Read an ADXL345 accelerometer over I²C, computed RMS and peak-to-peak vibration on the STM32, and streamed results over UART to a live Python dashboard.

RESEARCH FUNDING APPLICATION | MORE↗

ACE-FUELS (FUTO) — 2026–present. Official centre profile · Research-grant application in progress (not yet awarded), seeking support to further develop my first-author Human-on-the-Loop framework for fault-aware service restoration in Nigerian distribution feeders, with proposed work on the restoration model and staged simulation/validation protocol.

TECHNICAL SKILLS | MORE↗

Embedded	ARM Cortex-M (STM32, CubeIDE), ESP32 and ESP32-S3, ATmega328P, PIC; RTOS concepts; C and C++; UART, SPI, I²C, ADC, DMA
Hardware design	Altium Designer (schematics, multi-layer PCB, DFM/DRC), Proteus VSM, LTspice; analogue front ends, relay and driver circuits
Power and signals	Buck-boost and SMPS design, protection (overcurrent, earth fault), digital signal processing (DWT, FFT), control systems
Software and ML	MATLAB and Simulink, Python, Git, LaTeX, PlatformIO; TinyML with Edge Impulse

CERTIFICATIONS | MORE↗

- **Altium Certified** (Designer curriculum and cumulative exam), Sep. 2026
- **Arm Cortex-M Processors Overview** (Coursera), Jul. 2026
- **RTOS Concepts** (MindLuster), Jun. 2026
- **MATLAB and Simulink Onramp** (MathWorks), May 2024
- **Cisco Networking Academy**, Oct. 2023
- **Tait Basic Radio Awareness** (Tait Radio Academy)

CURRENT TRAINING PROGRAMMES AND FELLOWSHIPS | MORE↗

- Renesas / Altium Academic Programmes (PCB layout, DFM/DRC, stack-up); STMicroelectronics Ecosystem Academy (Cortex-M peripherals, low-power design, DMA); Texas Instruments Precision Labs (data converters, op-amp stability and signal integrity) — ongoing, Oct. 2026–present.

PROFESSIONAL AND TEACHING EXPERIENCE | MORE↗

Engineering Intern Jun. – Sep. 2025 (4 months)
Enugu Electricity Distribution Company (EEDC) / MainPower *Enugu, Nigeria*

- Rotated across seven departments; supported asset audits and load-profile calculations, including 100 kVA transformer sizing and an 11 kV fuse rating.
- Used Megger insulation- and earth-resistance testers; traced voltage harmonics to a failed damping resistor with a power-quality analyser and observed fault clearing and emergency load shedding.

Telecommunications Intern (part-time) Jul. – Nov. 2025
Federal Radio Corporation of Nigeria (FRCN) and Tait Radio Academy *Enugu and Abuja, Nigeria*

- Studied VHF/UHF propagation, repeater and trunked-radio design, and FDMA/TDMA modulation.

MATLAB Software Engineer (SIWES Intern) May 2024
Center for Social Awareness, Advocacy and Ethics (CSAAE)

- Wrote MATLAB scripts for numerical computing, data preprocessing and 2D/3D visualisation; simulated linear-algebra and differential-equation models in a six-person team.

Academic Assistant, Petroleum Engineering (100 Level, second semester) FUTO
Under class advisor Dr. Nnanna Okoli (a different department, to avoid grading bias)

- Graded assessments and maintained confidential result records under faculty supervision.

HONOURS, MEMBERSHIPS AND SERVICE | MORE↗

- Student Member, IEEE, and IEEE FUTO Student Branch. Departmental Letter of Academic/Character Commendation, Electronic Engineering, FUTO.
- Academic Coordinator / Organising Secretary, SU SchoolGroups, FUTO (2023–2025): coordinated peer tutorials for 200+ students and organised academic seminars.
- Skill-Up Imo Cohort 2 winner - Web App Development (2022); received a laptop award. **Photo↗**
- Languages: English (native; medium of instruction); Mandarin Chinese (elementary, HSK 1).